

**Lab 2 Report****Jose Montalvo**

```
cd "C:\Users\tesse\Documents\School\Winter 26\instrumentation"  
py ArduinoMindReader.py extra 9600 -v
```

## Page 1

What range of voltages can the Arduino Uno `analogRead()`?

Additionally, what range of values can `analogRead()` output on the Arduino Uno? (This corresponds to the number of values that can be described by an N-bit number: what is N?)

Find the specific page in the online Arduino documentation that provides this information and a) link to it in your answer

b) provide a screenshot of the appropriate paragraph with this information.

## Page 2

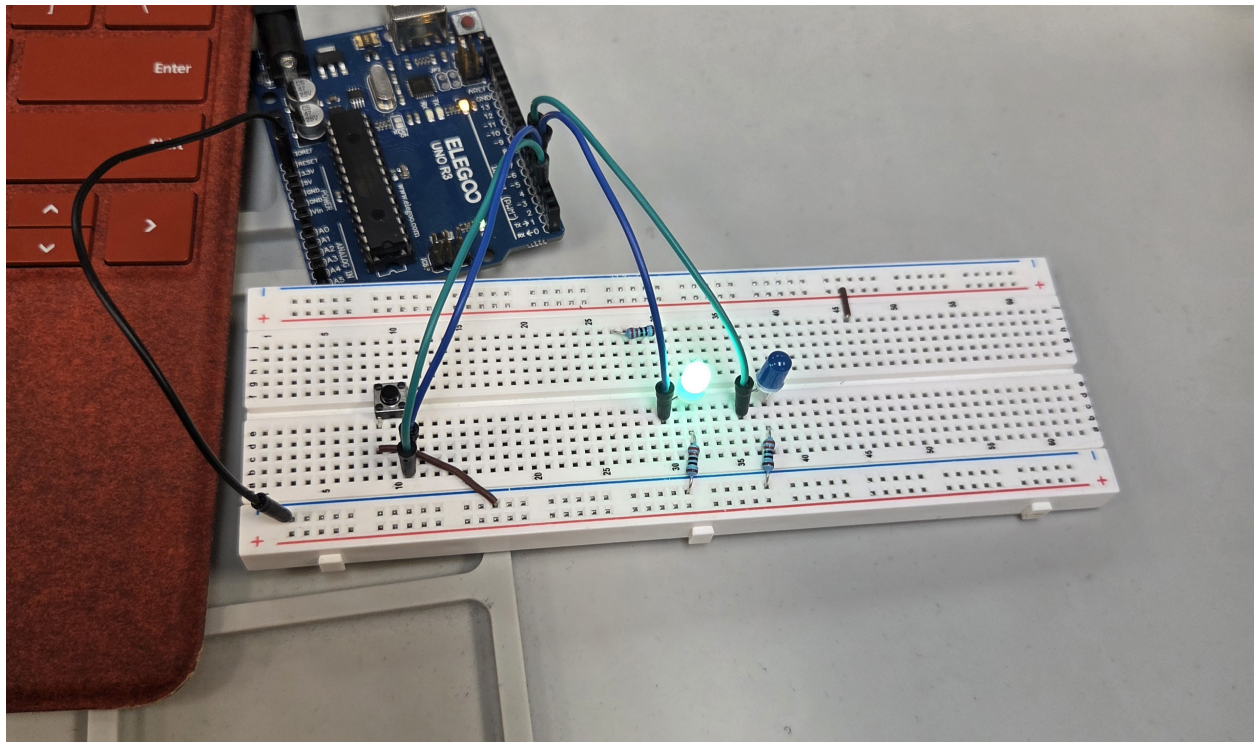


Figure 1: The interrupt will be continuously looking for the state change. In polling, you will only be notified if you hit at exactly the correct time.

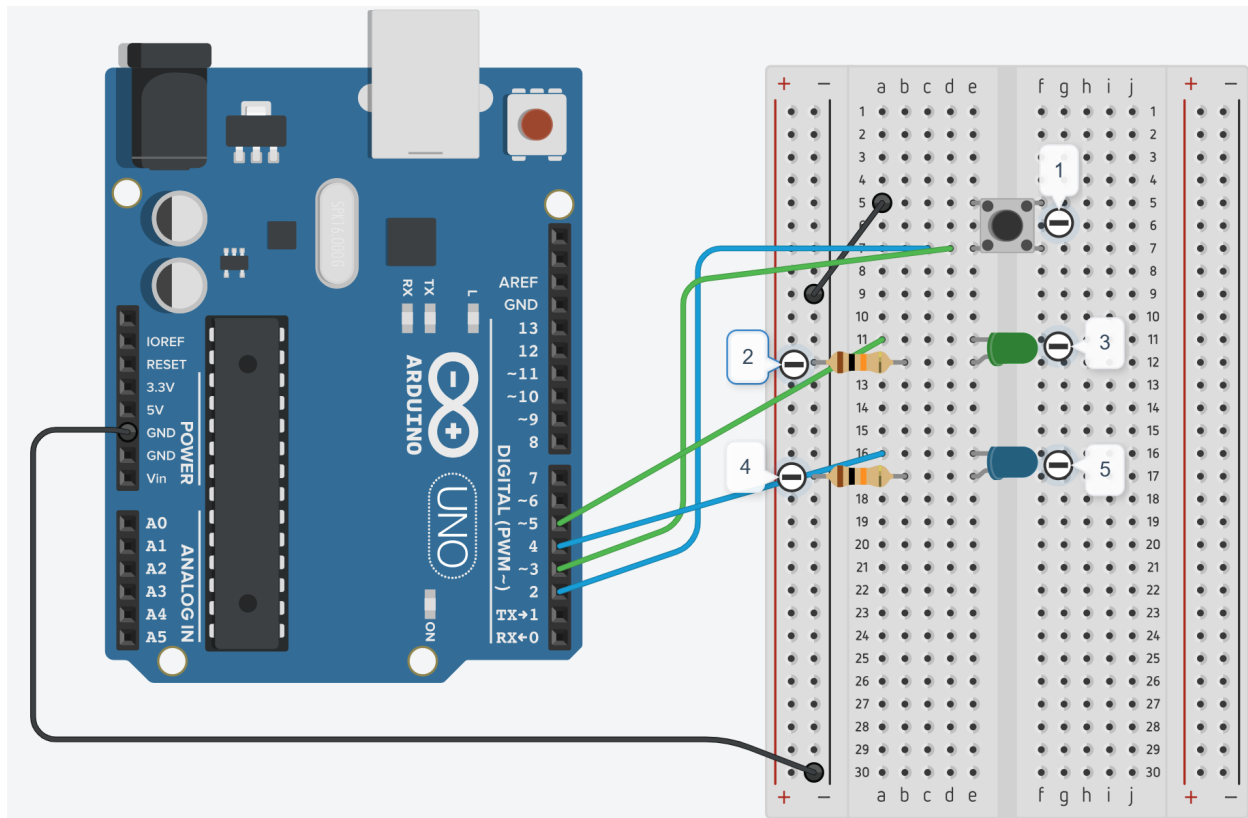


Figure 2: An interrupt does a better job at getting the button press over the polling since an interrupt is looking for changes of state continuously instead of the polling which is

| Component Number | Detail      |
|------------------|-------------|
| 1                | Button      |
| 2                | $R_1 = 10k$ |
| 3                | LED Green   |
| 4                | $R_2 = 10k$ |
| 5                | LED         |

Place your photo of the button racing circuit with only one lit LED in the center. Add a brief caption (1-3 sentences) that explains why an interrupt can capture the button press but polling might miss it.

## Page 3

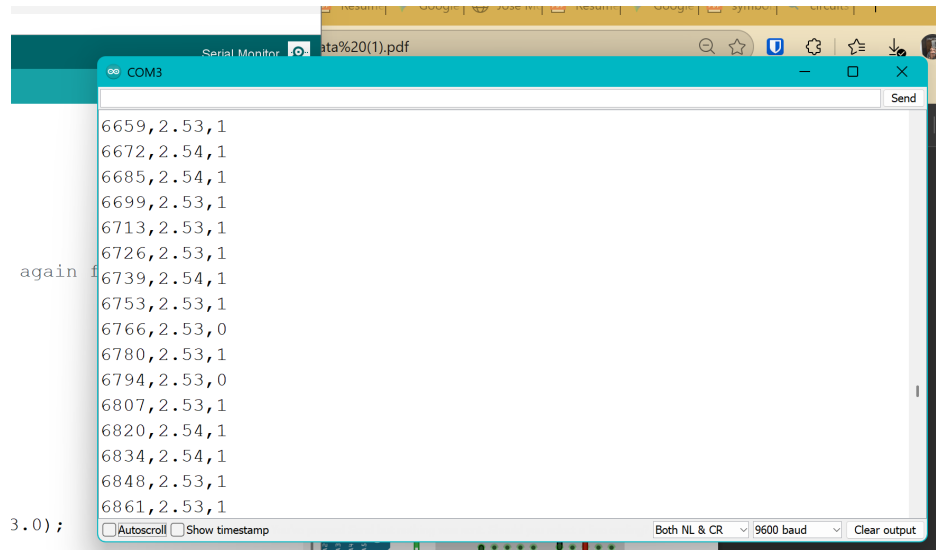


Figure 3: This data stream is showing the voltage values as I would turn the knob on the potentiometer. It seemed to be transitioning around 2.5ish[V] to go from low to high.

Place your screenshot of the digital value transitioning from 0 to 1 or vice versa in the center. Add a brief caption (1-3 sentences) describing what was being measured and what the voltage transition point seems to be between digital HIGH and LOW.

## Page 4

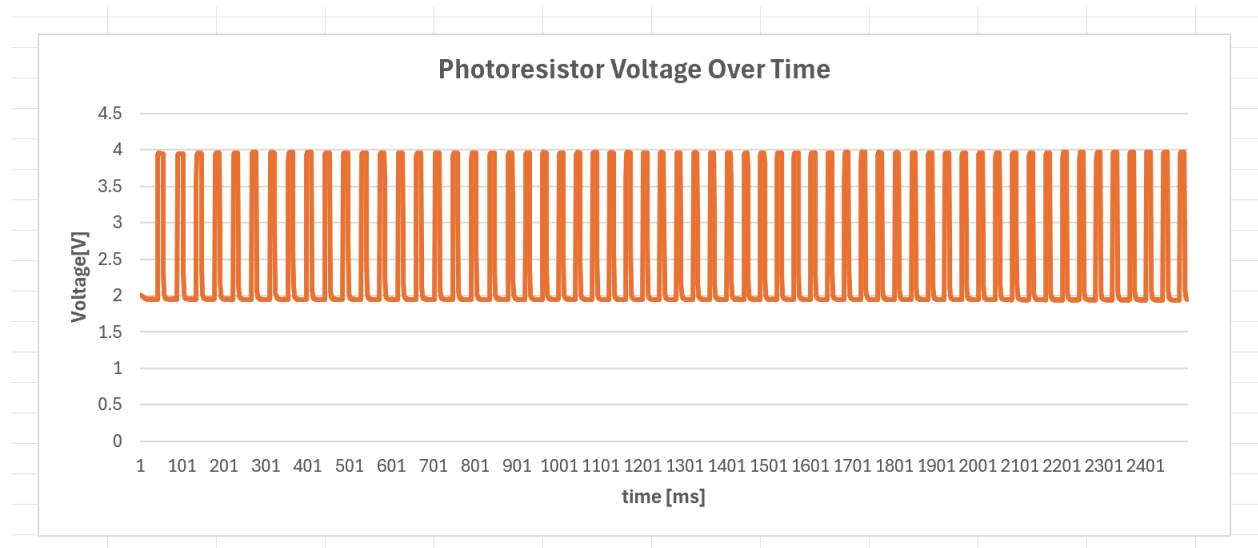


Figure 4: This plot is showing the pickup that the photoresistor was seeing. The output was a voltage and the freq was only working around 0.20 Hz on the app.

Place your plot of the light strobing data in the center. Add a brief caption (1-3 sentences) describing the plot, making sure to include the type of sensor used and the frequency of the light strobing.

## **Page 5**

No other sources

No Collaborators

## Page 6

To help students assume responsibility for their own learning in the course, we will request a brief self-evaluation in each student's own words. Please respond to the following prompts, minding the prescribed word limits.

1. What did you expect to learn during this lab? (Please respond in under 100 words.)

I expected to learn about basic digital and analog signal reading

2. How would you rate your attendance and participation in class during this lab experience? Are any adjustments needed in your approach for future labs, in order for you to be successful? (Please respond in under 100 words.)

I did a great job this week staying ahead. Getting the lab half done during the weekend is definitively helping.

3. How well did you do at the lab assignment itself? You might include thinking about strengths and weaknesses of your performance in this response. (Please respond in under 100 words.)

I did really well. Getting into interrupts again was a little rough but it came together fine.

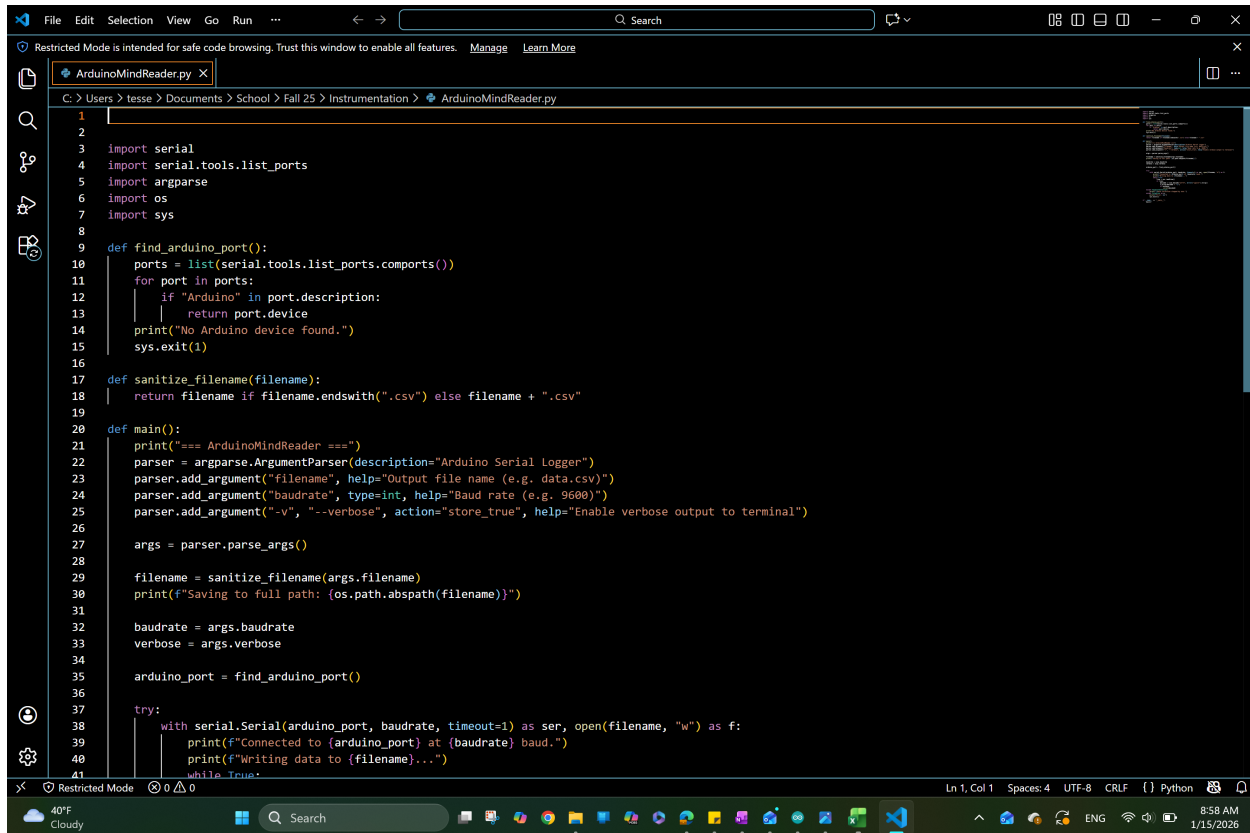
4. What did you actually learn in the lab? (This might include a range of things, from subject matter and concrete skills to ways of knowing and working together with peers.) (Please respond in under 100 words.)

I got more practice with reading and editing code. I always have a hard time getting back into that flow. One really nice thing was being able to get familiar with the variable types. That be something I struggle with in the future.



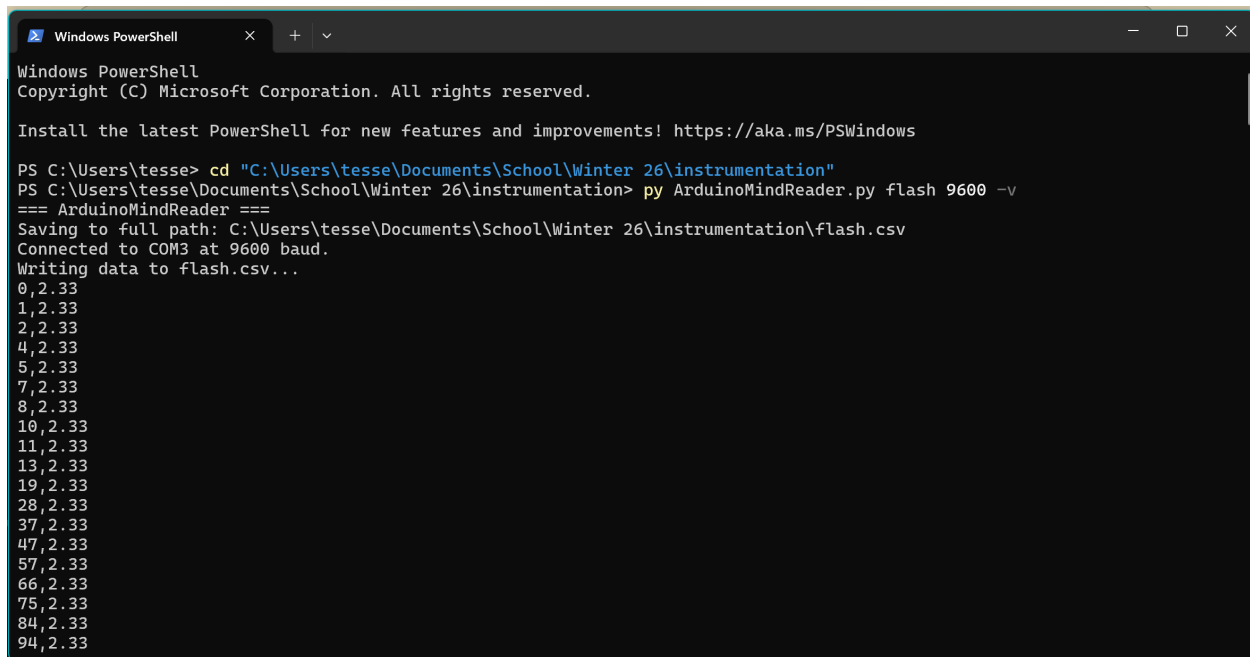
## Page 7

(optional, extra credit)



```
1
2
3 import serial
4 import serial.tools.list_ports
5 import argparse
6 import os
7 import sys
8
9 def find_arduino_port():
10     ports = list(serial.tools.list_ports.comports())
11     for port in ports:
12         if "Arduino" in port.description:
13             return port.device
14     print("No Arduino device found.")
15     sys.exit(1)
16
17 def sanitize_filename(filename):
18     return filename if filename.endswith(".csv") else filename + ".csv"
19
20 def main():
21     print("=== ArduinoMindReader ===")
22     parser = argparse.ArgumentParser(description="Arduino Serial Logger")
23     parser.add_argument("filename", help="Output file name (e.g. data.csv)")
24     parser.add_argument("baudrate", type=int, help="Baud rate (e.g. 9600)")
25     parser.add_argument("-v", "--verbose", action="store_true", help="Enable verbose output to terminal")
26
27     args = parser.parse_args()
28
29     filename = sanitize_filename(args.filename)
30     print(f"Saving to full path: {os.path.abspath(filename)}")
31
32     baudrate = args.baudrate
33     verbose = args.verbose
34
35     arduino_port = find_arduino_port()
36
37     try:
38         with serial.Serial(arduino_port, baudrate, timeout=1) as ser, open(filename, "w") as f:
39             print(f"Connected to {arduino_port} at {baudrate} baud.")
40             print(f"Writing data to {filename}...")
41             while True:
```

Figure 5: PythonIDE

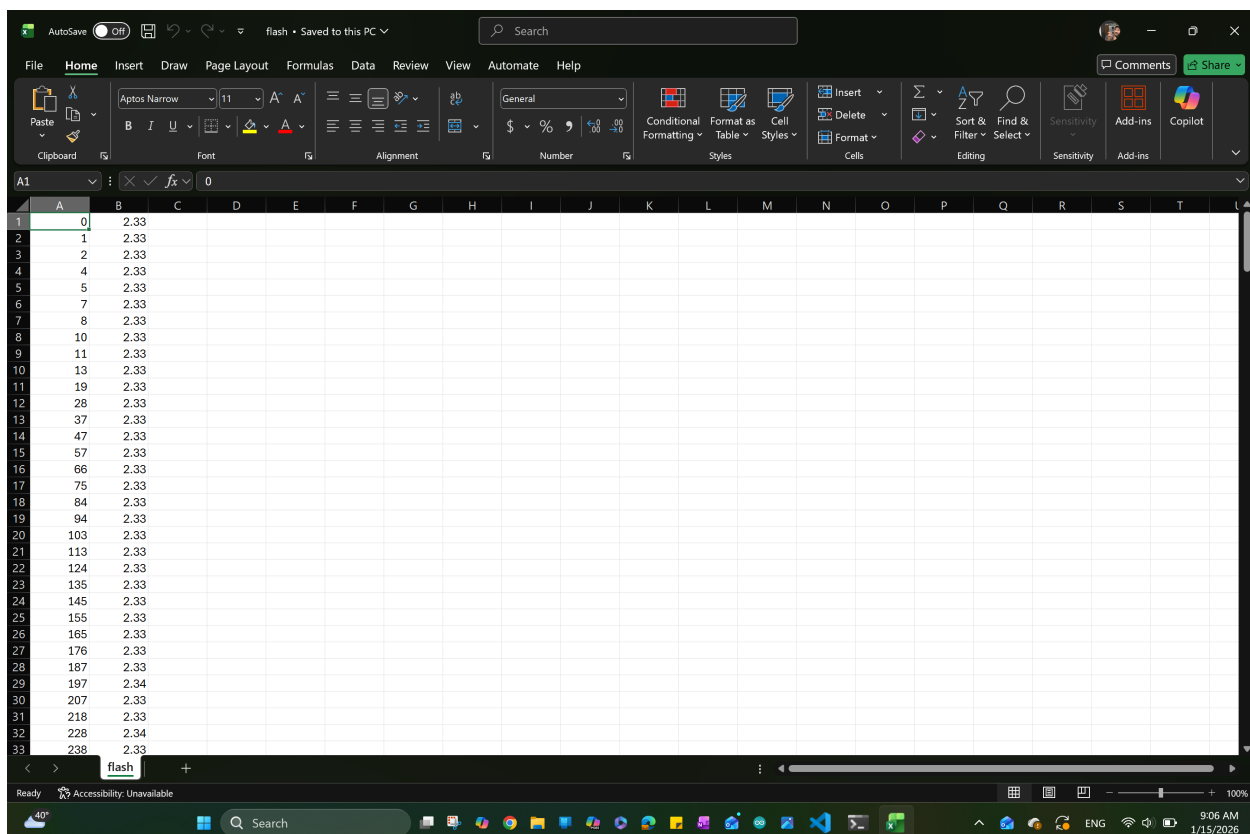


```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\tesse> cd "C:\Users\tesse\Documents\School\Winter 26\instrumentation"
PS C:\Users\tesse\Documents\School\Winter 26\instrumentation> py ArduinoMindReader.py flash 9600 -v
=== ArduinoMindReader ===
Saving to full path: C:\Users\tesse\Documents\School\Winter 26\instrumentation\flash.csv
Connected to COM3 at 9600 baud.
Writing data to flash.csv...
0,2.33
1,2.33
2,2.33
4,2.33
5,2.33
7,2.33
8,2.33
10,2.33
11,2.33
13,2.33
19,2.33
28,2.33
37,2.33
47,2.33
57,2.33
66,2.33
75,2.33
84,2.33
94,2.33
```

Figure 6: Start Command in Powershell



|    | A   | B    | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T |
|----|-----|------|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1  | 0   | 2.33 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 2  | 1   | 2.33 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 3  | 2   | 2.33 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 4  | 4   | 2.33 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 5  | 5   | 2.33 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 6  | 7   | 2.33 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 7  | 8   | 2.33 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 8  | 10  | 2.33 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 9  | 11  | 2.33 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 10 | 13  | 2.33 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 11 | 19  | 2.33 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 12 | 28  | 2.33 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 13 | 37  | 2.33 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 14 | 47  | 2.33 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 15 | 57  | 2.33 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 16 | 66  | 2.33 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 17 | 75  | 2.33 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 18 | 84  | 2.33 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 19 | 94  | 2.33 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 20 | 103 | 2.33 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 21 | 113 | 2.33 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 22 | 124 | 2.33 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 23 | 135 | 2.33 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 24 | 145 | 2.33 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 25 | 155 | 2.33 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 26 | 165 | 2.33 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 27 | 176 | 2.33 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 28 | 187 | 2.33 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 29 | 197 | 2.34 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 30 | 207 | 2.33 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 31 | 218 | 2.33 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 32 | 228 | 2.34 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 33 | 238 | 2.33 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |

Figure 7: Output file in excel

Screenshot of your Python IDE with your working script, terminal with command

```
python3 <your_script_name>.py extra 9600 -v
```

## Questions to think about

so I dont need to go looking for things come the final

1. AnalogRead Details

What range of voltages can the arduino in analogRead()? Can we change this range? If so how?

*analogRead() is able to read 0 to 1023 over a 5V range but we can change the voltage range to get a more appropriate resolution if we need it with analogReference()*

2. Debouncing

In this lab, we implemented software debouncing. Provide an example of a hardware debouncing circuit (RC version) and describe how it works. What are the pros and cons of using a hardware debouncing circuit?

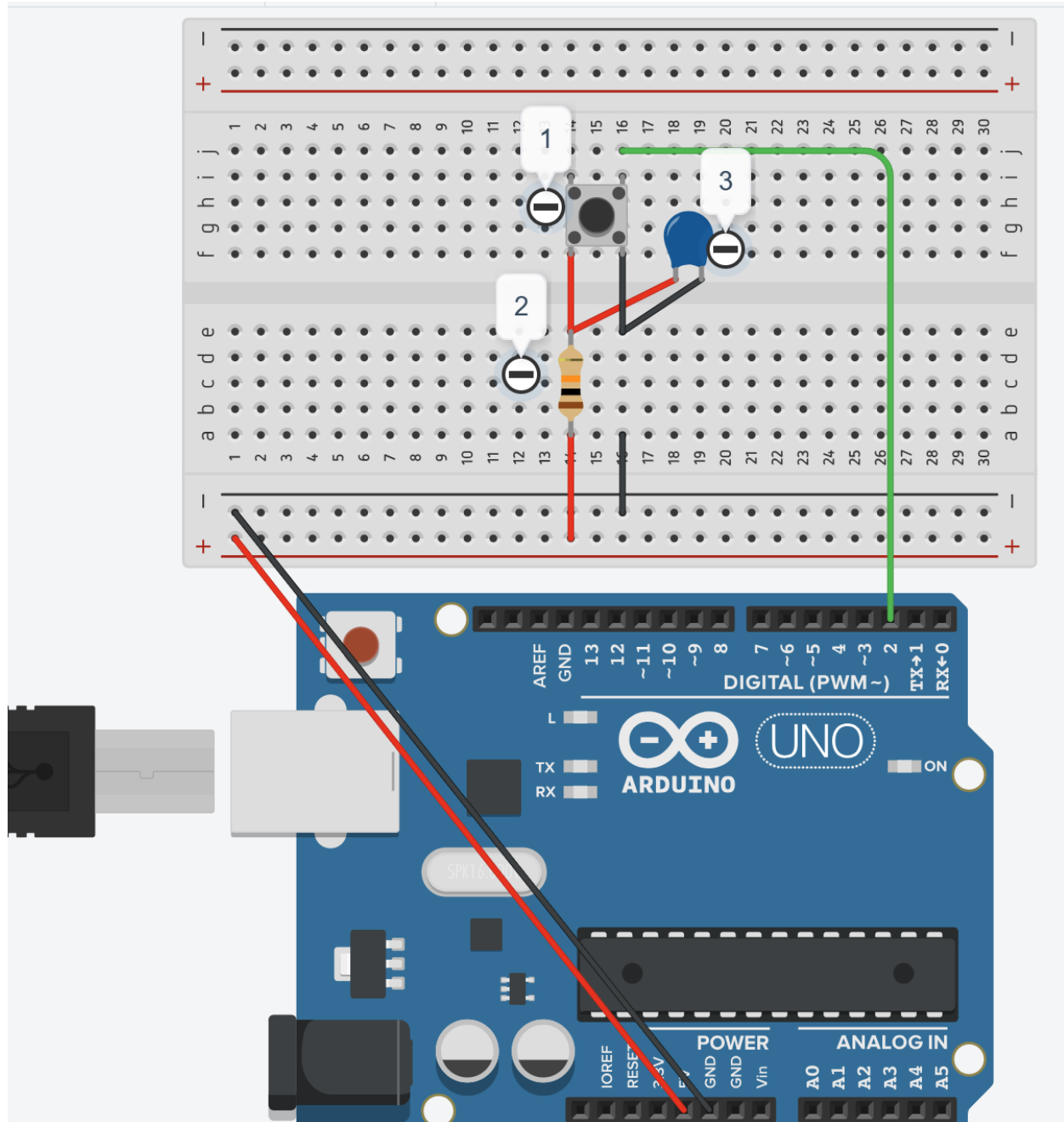


Figure 8: In this system, the capacitor will charge smoothly for an Unpressed button. When the button is pressed it will discharge to ground. In debouncing the capacitor acts as a filter.

### 3. Arduino Math

Most math in Arduino follows the basics:  $+$ ,  $-$ ,  $*$ , and  $/$ . However, there are some more complex cases. What do the following do? Please provide an example of you using it and the resulting value:

- i) `%`  
`int Tticks = 14;`

```
int stepsPerRev = 4;
int currentPosition = Tticks % stepsPerRev

ii) abs()
int setPoint = 100;
int currentVal = 120;
int errorMagnitude = abs(setPoint - currentVal);

iii) pow()
float radius = 3.0;
float area = 3.14 * pow(radius, 2);

iv) min()
int pwmCalc = 300;
int pwmMax = 255;
int motorOut = min(pwmCalc, pwmMax);

v) max()
int sensorVal = -5;
int floorVal = 0;
int cleanVal = max(sensorVal, floorVal);
```

**Code**

---

%CODE HERE

---